CHAPTER 2

Second degree: forms and choice of method

Chapter overview

Objectives. Choose a form; solve an equation or inequitable; read an extremum; demonstrate the root formula.
Prerequisites. Remarkable identities, zero product, product sign, square roots.
Annual benchmark. 22 September·1 October: 8 hours

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Develop ($x-3)^2$.

2. Factoring x^2-16.

3. Resolve ($x-2)(x+5)=0$.

4. Solve x^2 = 9 in $\mathbb{R}$.

5. Give the sign of 2 $x-6$.

6. Determine the minimum of ($x-1)^2+4$.

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Present two methods for factoring a trinomial.
All three cases were identified and explained why they were exhaustive.

Search before formalizing

A rectangle has an area of 24 m ^{2} and a perimeter of 20 m. Without randomly trying all the to determine its sides. Start with a drawing and explain the two constraints.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.

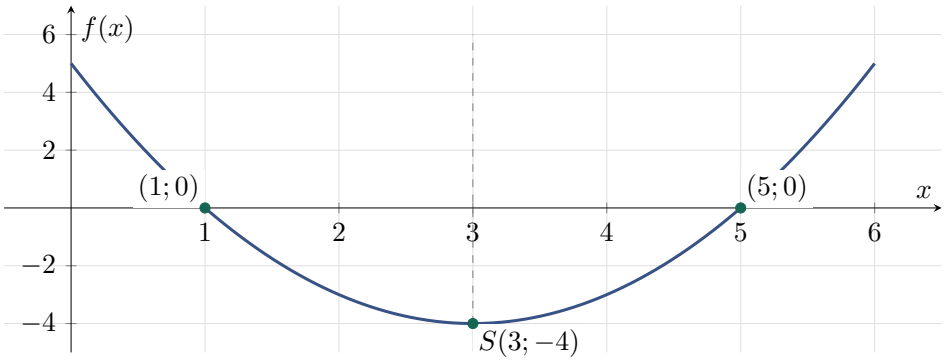
2.1 Three forms, three readings

Definition and results

A function of the second degree written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The developed form facilitates the calculation of $f(0)$; the canonical form $a(x-\alpha)^2 + \beta$ reveals summit and extremum; a factorized shape $a(x - r_1)(x - r_2)$ reveals roots and signs. Roots are the solutions of $f(x) = 0$.

Status and justification. The completion of the square is based on $x^2+2ux = (x+u)^2-u^2$. The existence of the canonical form is demonstrated by development; its general calculation is not an automatism payable.

Example solved. $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is -4 in 3 ; The roots are 1 and 5; the order originally is 5.



Curve of $f(x) = (x-3)^2 - 4$. The two roots and the top are indicated; the axis of symmetry is the right $x = 3$.

Point of vigilance

A factorized expression does not get by putting each of the terms separately between parentheses.

Immediate application. Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and sound minimum.

2.2 Discrimination and resolution

Definition and results

For $ax^2 + bx + c$ = 0 with $a \neq 0$, we put it down $b^2 - 4ac$. If < 0, no real root; if 0, one root $-\frac{b}{2a}$; if > 0, two roots $\frac{-b - \sqrt{\Delta}}{2a}$ and $\frac{-b + \sqrt{\Delta}}{2a}$. To order roots, consider the sign of a.

Status and justification. By multiplying the equation by 4 $a \neq 0$, one obtains (2 $ax + b)^2$ One. square is positive or zero: the three cases follow. > 0, $2ax + b = \pm \sqrt{\Delta}$. Disjunction of cases avoids take the square root of a negative number.

Example solved. $2x^2 + x - 3 = 0$: $\Delta = 1 + 24 = 25$; $x = \frac{-1 \pm 5}{4}$ i.e. 1 and $-\frac{3}{2}$. Verification: $(x - 1)(2x + 3) = 2x^2 + x - 3$.

Point of vigilance

The denominator is 2 a, not 2 ; the sign of b is taken with its parentheses in b^2.

Immediate application. Solve 3 $x^2 - 6x$ + 3 = 0 and $x^2 + 2x + 5 = 0$.

2.3 Factoring without automating the discriminant

Definition and results

If two real roots exist, their sum is $-\frac{b}{a}$ and their produce $\frac{c}{a}$. An apparent root or a remarkable identity often allows a shorter factorization. degree 2 cancelling in two distinct real r and s are exactly $a(x - r)(x - s)$ with $a \neq 0$.

Status and justification. Developing $a(x - r)(x - s) = ax^2 - a(r + s)x + ars$, identification the coefficients give the relationships. Conversely, if r and s have this sum and this product, the development regains the polynomial.

Example solved. $x^2 - 7x$ + 12 : 3 and 4 have sum 7 and produce 12. $x - 3)(x - 4$. For to impose more $f(0)$ = 24, it takes 12 a = 24, so $a = 2$.

Point of vigilance

The sum of the roots is $-\frac{b}{a}$; don't forget the sign less. Only one root is not enough to determine a.

Immediate application. Determine the level 2 f polynomial such that $f(1) = f(4)$ = 0 and $f(0) = -8$.

2.4 Sign, inequities and optimization

Definition and results

For two distinct roots $r < s$, the trinomial has the sign of a outside [$r; s$] and the sign between the roots. It is zero to the roots. Without real root, it keeps the sign of a. With a double root, it keeps the sign of a out of it. The optimization is done on the domain the problem, which may limit access to the summit.

Status and justification. The sign table of factors $a(x - r)(x - s)$ gives the result. canonical form compares $f(x)$ à β with the sign of $a(x - \alpha)^2$.

Example solved. $-2(x - 1)(x - 5) \geq 0$ equals $x \in [1; 5]$. $A(x) = x(12 - x)$ on [0; 12], $A(x) = 36 - (x - 6)^2 \leq 36$, with equality in 6. A rectangle of perimeter 24 m has therefore an area maximum of 36 m ^{2} when it is square.

Point of vigilance

In a strict inequation, the roots are excluded. Check the geometric bounds before conclude.

Immediate application. Solve $x^2 - 4x - 5 < 0$ then maximize $x(8 - x)$ on [0; 3].

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 2.1 · Complete a square

The function is considered f defined on $\mathbb{R}$ by $f(x) = x^2 - 8x + 7$.

1. Complete the square to write $f(x)$ in form ($x - 4)^2 + k$where: k is a real to be determined.
2. Deduce a factorization of $f(x)$ using a remarkable identity.
3. Draw up the signboard of f, then solve in $\mathbb{R}$ Inequation $f(x) \leq 0$.

Exercise 2.2 · Discuss without heavy calculation

Functions are studied $f(x) = x^2 + 4$, $g(x) = (x + 2)^2$ and $h(x) = -3(x - 1)(x + 4)$, all defined on $\mathbb{R}$.

1. For each function, determine any actual roots by directly exploiting the shape given.
2. Specify the sign of each function on $\mathbb{R}$. Indicate explicitly the values in which it cancels.
3. Explain why the calculation of a discriminant is not necessary in any of the three cases.

Exercise 2.3 · Selecting a Strategy

We wish to resolve in $\mathbb{R}$ the following three equations:

$$x^2 - 9 = 0, \qquad x^2 - 9x + 20 = 0, \qquad 2x^2 - 3x - 1 = 0.$$

1. For each equation, choose a method: remarkable identity, search for two numbers of sum and product known, or discriminating. Explain this choice.
2. Solve the three equations and give the solution sets in exact form.
3. Control the solutions of the first two equations by substitution. For the third, control the sum and product of the roots obtained.

Exercise 2.4 · Find a parabola

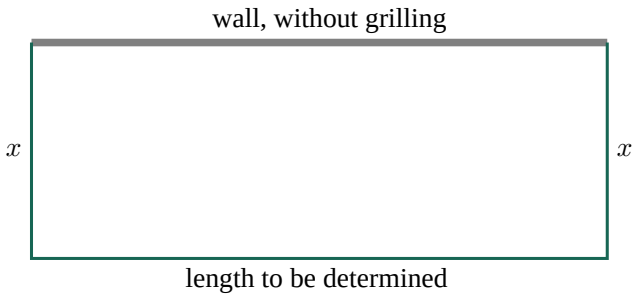
The curve of a polynomial function f of degree 2 cut the axis of the abscissas at the abscissa points -2 and 3. It also goes through the point $A(1; 12)$.

1. Determine an expression of $f(x)$ on the basis of this three information. Why she's okay.
2. Develop the term found. Deduce the order from the origin of the curve.
3. Verify that the function obtained satisfies the three initial informations.

Exercise 2.5 · A vegetable garden against a wall

A gardener has 20 m of grilling to surround three sides of a rectangular vegetable garden. fourth side runs along a wall and does not require grilling. x the length, in metres, of each of the two sides perpendicular to the wall. The whole length of the wire is used.

1. Make a diagram and express the length of the side parallel to the wall according to x. Specify the values of x for which the rectangle is not flattened.
2. To express the point $A(x)$, in square meters. Determine its maximum value and the corresponding dimensions, by choosing the method of your choice.
3. Check that these dimensions use exactly 20 m of grilling.



Vegetable garden diagram: only the three green lines use the 20 m of grilling. drawing does not fix the optimal dimensions.

Exercise 2.6 · A Parameter

For every real m, we consider the equation of actual unknown x :

$$x^2 - 2x + m = 0.$$

The number m is a parameter: it remains fixed when the equation is solved.

1. Determine the values of m for which the equation allows two distinct real solutions.

2. Determine the values of m for which it accepts a single real solution, then those for of which it does not admit any.

3. Give the solution in case it is unique. Explain why the three cases envisaged cover all values of m.

Exercise 2.7 · Link to the sequences

A sequence is defined by $u_n = n^2 - 5n + 4$ for all natural n. We're looking for all ranks to which the end of the sequence is strictly negative.

1. First solve in $\mathbb{R}$ Inequation $x^2 - 5x + 4 < 0$.

2. Deduce all natural integers n such that $u_n < 0$. Justify why a real interval does not This is not the final answer.

Fiscal year 2.8 · Deferred review

Student studying the function f defined on $\mathbb{R}$ by $f(x) = 2(x+3)^2 - 5$. She wrote: "The minimum value -5 and it is reached for $x = 3$. »

1. Indicate what is correct and what is wrong in this sentence. justifying your answer.

2. Solve in $\mathbb{R}$ Inequation $f(x) \leq -3$, then check the two bounds of the obtained set.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 43 p. 99, questions a and c. First find an obvious root, then the other with the help of the root product.

CQFD, exercise 49 p. 99, questions a and b. Translate sum and product by equation ; do not assume in advance that the numbers exist.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. The roots of ($x+1)(x-$ (4) are

   A. 1 and 4

   B. -1 and -4

   C. -1 and 4

   D. 1 and -4

2. The discriminant x^2-2x+3 is

   A. -8

   B. 8

   C. 16

   D. -2

3. The summit of $x-2)^2+5$ is

   A. $(2;-5)$

   B. $(-2;5)$

   C. $(5;2)$

   D. $(2;5)$

4. The sum of roots of 2 x^2-6x+1 is

   A. 6

   B. 3

   C. -3

   D. $\frac{1}{2}$

5. $x^2-4<0$ equals

   A. $x>2$

   B. $x<-2$

   C. $x\in]-2;2[$

   D. $x\in[-2;2]$

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. x^2+6x+9 a for factorization

   A. $(x-3)^2$

   B. $(x+9)(x+1)$

   C. $(x+6)^2$

   D. $(x+3)^2$

2. The root product of 3 x^2+x-2 is

   A. $-\frac{2}{3}$

B. $-\frac{1}{3}$

C. $\frac{2}{3}$

D. -2

3. $-x^2 + 4x$ reached on [0; 1] its maximum in

A. 2

B. 0

C. 1

D. 4

4. A double root requires

A. $a = 0$

B. $\Delta = 0$

C. $b = 0$

D. $c = 0$

5. Level 2 polynomials of roots 0 and 2 are

A. $a(x + 2)$

B. $a(x - 2)^2$

C. $x(x - 2)$ only

D. $ax(x - 2)$, $a \neq 0$

True or false?

1. A second degree function always has two real roots.

2. A trinomial having two distinct roots is negative between them.

3. For all real x, ($x - 2)^2 \geq 0$.

Personal review

I justify the choice of form; I distinguish roots and top; I end an inequitable by a
together; I respect the field of the model.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.

Corrected and conduct of chapter 2

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. $x^2 - 6x + 9$. *Action:* Reconstitute the product of two pairs.

2. $(x - 4)(x + 4)$. *Action:* Use the difference of two squares out of two examples.

3. x = 2 or $x = -5$. *Action:* Verbate the null product rule and its hypothesis.

4. $x = -3$ or $x = 3$. *Action:* Distinguish the positive square root from the equation solutions.

5. Negative before 3, zero in 3, positive after 3. *Action:* Draw the right and then make the signboard.

6. 4 reached only in 1. *Action:* From the positive character of a square.

Research activity

Positive lengths x and y check $x + y$ = 10 and xy = 24. They are the roots of $t^2 - 10t + 24 = (t - 4)(t - 6)$. Dimensions 4 m and 6 m; control of the area and perimeter.

Immediate applications

1. $g(x) = (x + 1)^2 - 4 = (x - 1)(x + 3)$. Roots 1 and -3, minimum -4 in -1.

2. The first is 3($x - 1)^2$ = 0, therefore 1. For the second one, $-16 < 0$: no real solution.

3. $f(x) = a(x - 1)(x - 4)$; $4a = -8$, so $f(x) = -2(x - 1)(x - 4)$.

4. $(x - 5)(x + 1) < 0$ gives] -1; 5[. The summit of $x(8 - x)$ is in 4, out of the domain; on [0; 3], the function grows and its maximum is 15 in 3.

Exercises: approach and results

Exercise 2.1 · Complete a square. $x^2 - 8x + 7 = (x - 4)^2 - 9 = (x - 1)(x -$ (7). The coefficient dominant is positive; the trinomial is negative or zero between the roots: [1; 7].

Exercise 2.2 · Discuss without heavy calculation. $x^2 + 4 > 0$ always, without root. ($x + 2)^2 \geq 0$, double root -2. For the last, roots -4 and 1 ; positive on] $-$ 4; 1[, negative on the outside, zero on the Boundaries.

Exercise 2.3 · Choose a strategy. Identity: $x = \pm 3$. Sum and product: 4 and 5.
third, no obvious whole root: $x = \dfrac{3 \pm \sqrt{17}}{4}$. The discriminant is suitable in
all cases, but is not always the most effective.

Exercise 2.4 · Find a parabola. $f(x) = a(x + 2)(x - 3$. In 1: $-6a$ = 12, therefore $a = -2$; $f(x) = -2x^2 + 2x$ + 12, where $f(0) = 12$.

Exercise 2.5 · A vegetable garden against a wall. The two widths use 2 x meters; length $20 - 2x$. Domain [0; 10] for the function,]0; 10[for an undegenerated rectangle. $A(x) = 20x - 2x^2 = 50 - 2(x - 5)^2$. Maximum 50 m ^{2}dimensions 5 m and 10 m.

Exercise 2.6 · A parameter. $\Delta = 4 - 4m$. Two roots if $m < 1$; double root 1 if $m = 1$; none if $m > 1$. The parameter m remains fixed when resolving in x.

Exercise 2.7 · Link to the sequences. $u_n = (n - 1)(n - 4)$; negative for 1 $< n <$ 4. Like n is integer, the only rows are 2 and 3. All ranks are not the integer real range.

Exercise 2.8 · Deferred review. The minimum is -5 in -3. The equation gives ($x + 3)^2 \leq 1$, let $-1 \leq x + 3 \leq 1$, therefore $[-4; -2]$.

CQFD, exercise 43 p. 99, questions a and c. a: obvious root 1, product -30, so second root -30. c: write 4 $x^2 + 9x$ + 5 ; root -1, product $\frac{5}{4}$, so second root $-\frac{5}{4}$. Check the amounts, respectively -29 and $-\frac{9}{4}$.

CQFD, exercise 49 p. 99, questions a and b. Numbers are the roots of $t^2 - st + p$. a : $t^2 - t$ + 1, discriminant -3 : no actual torque. b : $t^2 + 60t$ + 756, discriminant 576 = 24 ^{2} : -18 and -42. Their sum is equal to -60 and their products 756.

MCQ: answers and reasons

MCQ 1.

1. C. Cancel each factor.

2. A. $4 - 12 = -8$.

3. D. The square cancels in 2, value 5.

4. B. $-\frac{b}{a} = \frac{6}{2} = 3$.

5. C. Positive dominant coefficient, strict inequality.

MCQ 2.

1. Remarkable identity.

2. A. $\frac{c}{a} = -\frac{2}{3}$.

3. C. The function grows up to 2; on this domain, right edge 1.

4. B. For non-zero, the null discriminant characterizes the double root.

5. D. The multiplier coefficient is free and not zero.

True or false

1. False: x^2 + 1 is strictly positive.

2. False without hypothesis on a; $-(x - 1)(x - 2)$ is positive between 1 and 2.

3. True: the square of a real is positive or null; equality only in 2.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.